

SchNet Training Report

Phase 1 Baseline Bake-off — Sprint 3 (SCRUM-53)

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Summary

SchNet — continuous-filter convolutions over Gaussian-expanded interatomic distance (Schütt et al., 2017) — was trained to predict molecular energy and interatomic forces for MD17 ethanol at the DFT level of theory, as one of five architectures in the Phase 1 baseline bake-off (alongside MPNN, PaiNN, DimeNet++, and PhysNet). This is currently the only entry in that bake-off trained for the full epoch budget on the complete shared dataset, making it the project's reference Phase 1 result as of this report.

Setup

Data	data/gold/md17/ethanol_dft.npz — 9 atoms, 555,092 configurations
Split	Fixed 80/10/10 contiguous trajectory blocks, baked into the gold file — deterministic, no randomness, identical for every bake-off entry
Architecture	hidden_channels=64 , num_layers=3 , num_rbf=16 , cutoff=5.0 Å
Optimiser	Adam, lr=5e-4 , weight_decay=1e-6 , gradient clipping (max_norm=10), ReduceLR0nPlateau
Batch size / seed	256 / 42
Loss weighting	energy_loss_weight=1.0 , force_loss_weight=100.0 — fixed across all five bake-off models
Code	ml/models/schnet.py (model) + ml/training/train_schnet.py (standalone train/eval script)

Key implementation decision: energy centering

Ethanol's raw total energy sits at approximately **-97,195.9 kcal/mol, with a standard deviation of only ~4.2 kcal/mol** — the physically meaningful signal is a small variation riding on an enormous constant. Training directly against the raw value would spend most of early optimisation simply learning that constant offset rather than anything physically useful. Instead, the model is trained against `E - mean(train_E)`, with the shift added back before any metric is computed — so every number reported below is a genuine physical quantity, directly comparable to the other four architectures in the bake-off, not an artefact of this internal technique. The shift value (`-97195.891`) is stored inside the saved checkpoint for reproducibility.

A second, unrelated correctness issue was identified and corrected during this work: the original prototype this project's MPNN baseline was drafted from computed forces from a precomputed edge feature with no gradient connection back to atomic positions — meaning `torch.autograd.grad` would have nothing to differentiate, and forces would silently come out as exact zeros with no error raised. SchNet instead recomputes the relevant distance features from the live atomic positions inside the force-computation path, and this was verified correct with a finite-difference check (0.004008 by finite difference vs. 0.003984 by autograd, on a controlled synthetic test) before any training on real data began.

Training behaviour

The model was trained for the full 200-epoch budget and never triggered early stopping (patience of 20 epochs) — validation loss continued to find small improvements throughout the run. Train and validation loss tracked each other closely for the entire run, rather than diverging, which is the expected signature of a model that is still learning rather than overfitting. The learning rate was reduced automatically four times by the scheduler as progress plateaued ($5e-4 \rightarrow 2.5e-4 \rightarrow 1.25e-4 \rightarrow 6.25e-5 \rightarrow 3.13e-5$), with each reduction producing a further small drop in loss. The metrics reported below use the weights from epoch 198 — the epoch with the best validation loss — rather than epoch 200, since the final two epochs were both marginally worse.

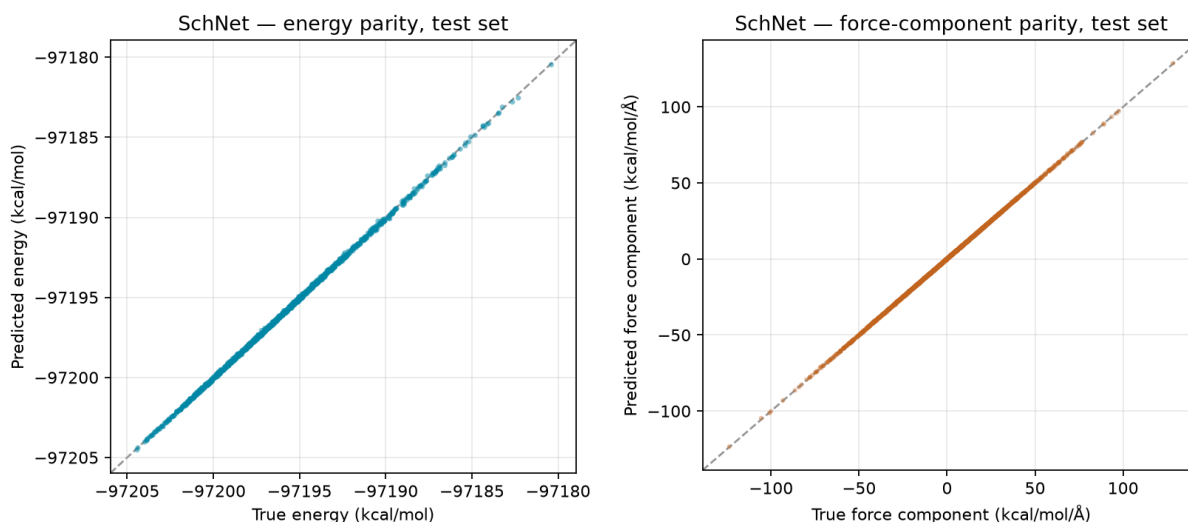
Results

Evaluated on the held-out test split (55,509 frames, 499,581 atoms), untouched until this final evaluation, scored via the project's shared metric harness (`ml.utils.metrics.MetricAccumulator`), so these numbers are directly comparable to the other bake-off entries without unit or convention caveats.

Metric	Value
Energy MAE	0.0873 kcal/mol
Energy RMSE	0.1026 kcal/mol

Metric	Value
Energy MAE (per atom)	0.0097 kcal/mol/atom
Force MAE (component)	0.1114 kcal/mol/Å
Force RMSE (component)	0.1628 kcal/mol/Å
Force MAE (atom-norm)	0.2231 kcal/mol/Å
Force RMSE (atom-norm)	0.2820 kcal/mol/Å
Training time	30,029 s (~8h 20m), CPU
Inference time	9.03 s total / 0.163 ms per frame, full test set

The two charts below, regenerated directly from the saved checkpoint against the real test set, show how tightly the model's predictions track the true values — each point close to the dashed line indicates a correct prediction.



Force predictions show more spread than energy, which is expected — force is the harder quantity to predict, since it is a per-atom, per-direction derivative rather than a single scalar per molecule.

Reproducibility

- Checkpoint: `experiments/checkpoints/phase1_schnet.pt` (not committed — gitignored by project convention; regenerate via `python -m ml.training.train_schnet`).
- Results: `experiments/results/phase1_schnet.json` (also gitignored).
- Configuration: `experiments/configs/phase1_schnet.yaml`.
- Fixed seed (42) and a deterministic data split mean a re-run should reproduce these results closely, floating-point non-determinism aside.

Known limitations

- Trained and evaluated on ethanol at the DFT level only — not yet run against the other MD17 molecules or the larger MD22 systems.
- `ml/training/train_schnet.py` is a temporary, architecture-specific script. Once the shared `ml/training/train.py / evaluate.py` harness is complete for every architecture, this result should be reproduced through it, so that every bake-off entry has gone through identical code as well as identical data.
- This is a deterministic Phase 1 baseline with no uncertainty quantification — that is explicitly Phase 2/3 scope, not a gap in this entry.

Context within the Phase 1 bake-off

As of this report, this is the only bake-off entry trained on the full shared dataset for the full epoch budget. MPNN's shared training harness exists but has only been smoke-tested; DimeNet++ was trained on a different (CCSD(T)) dataset; PhysNet was trained on a 4.5% data subset for compute reasons; PaiNN is implemented but not yet trained. This result is therefore the project's current reference point for what a fully converged Phase 1 baseline looks like, pending the remaining four entries being brought onto the same footing.